
Problem Set 3: Complex Numbers

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Extra Problem

Clearly, $\mathcal{S}$ is just $\mathbb{Z}[i]$. I don't feel like proving this the annoying way so I will simply show $\mathbb{Z}[i]$ is isomorphic as a ring to $\mathbb{Z}[x]/(x^2 + 1)$, which is guaranteed to be a ring since $(x^2 + 1)$ is an ideal. Note that there exists a group isomorphism

$$\frac{\mathbb{Z}[x]}{(x^2 + 1)} \cong_{\mathbf{Grp}} \mathbb{Z} \oplus \mathbb{Z}$$

because $f(x) := x^2 + 1 \in \mathbb{Z}[x]$ has degree 2. Consider the definition of multiplication in $\mathbb{Z}[x]/(f(x))$. Since f is monic, and since $\mathbb{Z}$ is a (idk where to go from here i think f just needs to divide all degree ≥ 2 polynomials)